

Supplemental Material: Effective Perturbation Budget: Per-Seed Detail Tables

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TABLE S1

PER-SEED PAIRED COMPARISON OF DEFENDED-TRAINED PPO AND STATIC BETA ON THE EPB-BOUNDED DISTRIBUTION. $D = \text{PPO} - \text{BETA}$ IS THE PAIRED DIFFERENCE IN STANDARDIZED-UNIT MEAN TARGET-SCORE DEGRADATION.

Dataset	Seed	BETA	PPO	D
WADI	0	-0.08	+0.31	+0.39
WADI	1	-0.12	+0.41	+0.53
WADI	2	-0.09	+0.62	+0.71
WADI	3	-0.07	-0.04	+0.03 [†]
WADI	4	-0.06	+0.38	+0.44
SWaT	0	-0.10	+0.28	+0.38
SWaT	1	-0.09	+0.35	+0.44
SWaT	2	-0.08	-0.07	+0.01 [†]
SWaT	3	-0.07	+0.51	+0.58
SWaT	4	-0.05	+0.69	+0.74
paired mean		-0.081	+0.341	+0.42

[†] PPO non-convergence within the 200K-step budget.

This supplement collects the per-seed and per-detector detail tables underlying the aggregate results of the main paper (Audit, Defense, and Adaptive Validation). All conclusions drawn from these tables are stated in the main text. Unless a table quotes the BETA paper explicitly, every BETA result refers to our independent reimplementations (main text, Section V-A). Adaptive Validation: Table S1 (per-seed PPO-vs-BETA degradation), Table S2 (per-seed IPAL F1), Table S4 (ϵ -sweep), Table S5 (decision-level attacker configuration), Table S6 (query-based black-box attackers). Audit generality: Table S8 (per-seed EPB), Table S9 (normalization invariance), Table S10 (non-GNN detector). Defense: Table S7 (degradation reduction), Table S3 (val-99.5 FPR), Table S11 (calibration-split gap), Table S12 (systems latency percentiles), Table S13 (clipping side-effects by window slice).

Decision-level attacker specification: The decision-level attacker (main text, Section VII) minimizes the operator-facing top- k aggregate score $S_k(g(X+\delta))$ directly, observing only that scalar (no gradient, no surrogate, no per-sensor label). It runs in three stages per window: (i) *budget selection* — probe each of the N sensors by bumping it $+\epsilon$ and rank by the resulting S_k reduction, keeping the top- b as the perturbed set $\bar{V}$ (N queries); (ii) *random search* — for `iters` iterations sample a signed displacement $\delta_{\bar{V}} \sim \mathcal{U}[-\epsilon, \epsilon]^{b \times W}$ on $\bar{V}$, project by the deployed clip, query S_k , and keep the incumbent iff S_k strictly decreases; (iii) *sign refinement* — try $\pm\epsilon$ on each budget sensor's full window ($2b$ queries). The threshold is *not* used during search (it is applied only post-hoc to score

TABLE S2

PER-SEED IPAL F1 AT THE F1-OPTIMAL THRESHOLD PROTOCOL ON THE DEFENDED DETECTOR, CLEAN / BETA-ATTACKED / PPO-ATTACKED.

Dataset	Seed	Clean def	BETA def	PPO def
GDN-WADI	0	0.425	0.302	0.272
	1	0.473	0.406	0.473
	2	0.323	0.323	0.322
	3	0.368	0.363	0.368
	4	0.310	0.310	0.312
	mean	0.380 ± 0.062	0.341 ± 0.039	0.349 ± 0.069
GDN-SWaT	5-seed range	0.501–0.752	0.500–0.752	0.501–0.752
	mean	0.657 ± 0.102	0.658 ± 0.102	0.658 ± 0.102
TopoGDN-WADI [†]	4-conv. seeds	0.411–0.429	0.000–0.416	0.365–0.426
	mean	0.417 ± 0.008	0.137 ± 0.164	0.402 ± 0.025

[†] TopoGDN-WADI: seeds {0, 1, 2, 4}; seed 3's PPO did not converge within the 200K-step budget.

TABLE S3

EMPIRICAL CLEAN-DATA FPR OF THE VAL-99.5-PERCENTILE THRESHOLD HEURISTIC ON THE DEFENDED DETECTOR, 5 SEEDS PER CONFIGURATION. THE HEURISTIC IS A NO-LABELED-HOLDOUT FALLBACK; ACROSS ALL 20 SEEDS THE FPR IS AT LEAST 62%, INDICATING THE VAL-99.5 HEURISTIC IS NOT A DEPLOYABLE SUBSTITUTE FOR THE CAL-SPLIT PROTOCOL ON THE FOUR CONFIGURATIONS EVALUATED.

Configuration	FPR mean	FPR range	Spread
GDN-WADI	65.7%	62.75–67.48	1.1×
GDN-SWaT	73.8%	62.77–79.13	1.3×
TopoGDN-WADI	69.0%	62.40–76.98	1.2×
TopoGDN-SWaT	100.0%	100.00–100.00	1.0×

evasion), parameters are fixed a priori (not tuned on test windows), and WADI and SWaT use identical $b, k, \epsilon, \text{iters}$. Table S5 lists the settings.

TABLE S4

ε -SWEEP ON DEFENDED GDN-WADI (3 SEEDS, 30 TRIALS PER SEED). BETA UNDEF. DEGRADATION IS CONSTANT AT $\approx 32,862$ BECAUSE THE OOR-DRIVEN AMPLIFICATION FACTOR IS INDEPENDENT OF ε ONCE ε EXCEEDS THE IN-RANGE PERTURBATION SCALE.

ε	PPO def.	BETA def.	BETA undef.	BETA red.
0.01	0.067	0.001	32,862	$2.5 \times 10^7 \times$
0.05	0.281	0.007	32,862	$4.6 \times 10^6 \times$
0.10	0.431	0.015	32,862	$2.2 \times 10^6 \times$
0.20	0.599	0.033	32,862	$9.9 \times 10^5 \times$
0.50	0.726	0.219	32,862	$1.5 \times 10^5 \times$

TABLE S5

DECISION-LEVEL ATTACKER CONFIGURATION (GDN, DEFENDED PIPELINE). QUERY COUNTS ARE PER WINDOW; THE MEDIAN TOTAL IS REPORTED OVER ATTACKED WINDOWS. THE ATTACK IS PER-WINDOW AND OPERATES ON A LOGGED WINDOW (NON-CAUSAL).

Parameter	Value
Objective	$\min_{\delta} S_k(g(X+\delta)), k=10$
Norm / budget	$\ell_{\infty}, \varepsilon=0.1$
Sensor budget b	5
Budget selection	greedy $+\varepsilon$ probe, top- b (N queries)
Random-search iters	300 (matched), 400 (curve)
Sign refinement	$\pm\varepsilon$ per budget sensor ($2b$ queries)
Restarts	1 (seeded per window)
Initialization	$\delta=0$ (incumbent = clean)
Acceptance rule	keep iff S_k strictly decreases
Observation	scalar S_k only
Threshold access	none during search
Stopping	fixed query budget
Median queries/window	438 (WADI) / 362 (SWaT)
WADI vs. SWaT	identical b, k, ε , iters

TABLE S6

GRADIENT-FREE BLACK-BOX ATTACKERS ON THE DEFENDED GDN DETECTOR, F1-OPTIMAL THRESHOLD, MEAN OVER 5 SEEDS PER DATASET. “CONT. DEG.”: CONTINUOUS TARGET-SCORE DEGRADATION. “F1 RED.”: REDUCTION IN RAW F1. NO BLACK-BOX ATTACKER REDUCES F1, ON ANY SEED.

Dataset	Attacker	Cont. deg.	F1 red.
WADI	Random search	+8.84	0.000
	SPSA	+6.39	0.000
SWaT	Random search	+5.91	0.000
	SPSA	+4.67	0.000

TABLE S7

BETA MEAN TARGET-SCORE DEGRADATION BEFORE/AFTER THE INTERVENTION (MAIN TEXT, EQ. (4)), GDN ONLY. “UNDEF. (BETA, AS REPORTED)” QUOTES THE BETA PAPER (XAVIAR AND ARDAKANIAN, ARXIV:2509.17987V1); “DEF. (OURS)” IS OUR DEFENDED PER-SEED RESULT ON OUR REIMPLEMENTATION ($n=5$). “REDUCTION” IS THE RATIO OF THE TWO COLUMNS AND IS *conservative*: OUR OWN UNDEFENDED REPRODUCTION IS LARGER (5-SEED GDN-WADI MEAN 54,640), SO THE REDUCTION MEASURED ON OUR REPRODUCTION IS LARGER STILL. TOPOGDN ROWS ARE OMITTED (OUR 5-SEED UNDEFENDED MEANS ARE 66,968 ON WADI AND 1,081 ON SWaT); THE TOPOGDN-WADI OUTCOME IS REPORTED IN MAIN-TEXT SECTION VI-C.

Config	Undef. (BETA, as reported)	Def. (ours)	Reduction
GDN-WADI	$\approx 18,900$	~ 0.1	$\sim 1.9 \times 10^5 \times$
GDN-SWaT	≈ 146	~ 0.1	$\sim 1.5 \times 10^3 \times$

TABLE S8

PER-SEED EPB STATISTICS ON GDN, $n=50$ TRIALS PER SEED, AT NOMINAL $\varepsilon=0.1$. THE CROSS-SEED VARIANCE IS SMALL BECAUSE THE OOR-CLAMP CONSTRUCTION IS DETERMINISTIC IN THE OUT-OF-RANGE MAGNITUDE OF THE INPUT WINDOW.

Config	Seed	$\bar{\varepsilon}_{\text{eff}}$	$\bar{\varepsilon}_{\text{eff}}^{\text{max}}$	Ratio
GDN-WADI	1 (headline)	261.90	267.29	2,619 $\times$
GDN-WADI	5-seed mean	260.7	267.1	$\sim 2.6 \times 10^3$
GDN-SWaT	0 (headline)	2.80	3.03	27.99 $\times$
GDN-SWaT	5-seed mean	2.78	3.02	$\sim 28 \times$
FGSM	GDN-WADI s0	0.09	0.10	$\sim 0.9 \times$

TABLE S9

OUT-OF-RANGE TEST DRIFT UNDER THREE TRAIN-FIT NORMALIZATIONS (NO DETECTOR, NO ATTACK). “% OOR”: FRACTION OF TEST CELLS OUTSIDE THE TRAIN-FIT RANGE. “MAX DRIFT”: MAXIMUM TEST DRIFT BEYOND THE TRAIN RANGE AS A MULTIPLE OF THE TRAIN-RANGE WIDTH. VALUES ARE IDENTICAL ACROSS SCHEMES BECAUSE ALL THREE ARE AFFINE PER-SENSOR MAPS.

Dataset	Normalization	% OOR	Max drift
WADI	MinMax	0.49%	527 $\times$
	z-score	0.49%	527 $\times$
	robust (med/IQR)	0.49%	527 $\times$
SWaT	MinMax	10.1%	113 $\times$
	z-score	10.1%	113 $\times$
	robust (med/IQR)	10.1%	113 $\times$

TABLE S10

DETECTOR-AGNOSTIC OOR-CLAMP AUDIT: GDN VS. A NON-GNN LSTM FORECASTER ON 2,000 OUT-OF-RANGE WINDOWS. “EPB”: INPUT DISPLACEMENT FROM THE CLAMP (DETECTOR-INDEPENDENT). “OOR-CLAMP DEG.”: UNCLIPPED-BASELINE MINUS CLIPPED-ATTACKED SCORE AT THE TARGET SENSOR. “REPAIRED”: SAME UNDER SYMMETRIC CLIPPING.

Dataset	Detector	EPB	OOR-clamp deg.	Repaired
WADI	GDN	161.7	+11,414	+0.000
	LSTM (non-GNN)	161.7	+12,140	+0.000
SWaT	GDN	0.82	+12.5	+0.000
	LSTM (non-GNN)	0.82	+9.3	+0.000

TABLE S11

CAL-SPLIT F1 VS. F1-OPTIMAL ORACLE, 20 (ARCHITECTURE $\times$ DATASET $\times$ SEED) CONFIGURATIONS. EACH ROW’S MAX GAP IS OVER 5 SEEDS $\times$ 5 RANDOM CAL/EVAL SPLITS PER SEED. CROSS-CONFIG MAX GAP IS 0.012; MEAN GAP IS +0.0009.

Config	$ F1_{\text{cal}} - F1_{\text{opt}} _{\text{max}}$	Mean gap	N
GDN-WADI	0.0115	-0.0003	5
GDN-SWaT	0.0060	+0.0007	5
TopoGDN-WADI	0.0119	+0.0049	5
TopoGDN-SWaT	0.0067	-0.0019	5

TABLE S12

SYSTEMS LATENCY, SINGLE CPU THREAD (APPLE M4), 1000 TIMED RUNS AFTER 100 WARM-UP, PER WINDOW (BATCH 1). THE CORRECTION ADDS NEGLIGIBLE OVERHEAD; THE EPB AUDIT IS OFFLINE.

Component	median (ms)	p95	p99
<i>WADI</i> ($N=127$, $W=100$)			
Baseline preprocess	0.0011	0.0012	0.0013
Corrected preprocess	0.0014	0.0015	0.0016
EPB audit / trace	0.0088	0.0100	0.0249
Detector inference	2.043	2.520	3.143
End-to-end corrected	2.090	2.538	2.848
<i>SWaT</i> ($N=51$, $W=100$)			
Corrected preprocess	0.0010	0.0010	0.0010
Detector inference	0.359	0.453	0.649
End-to-end corrected	0.372	0.514	0.980

TABLE S13

CLIPPING SIDE-EFFECTS: DETECTION BY WINDOW SLICE (GDN, SEED 0) AT A 5%-FPR BENIGN-CALIBRATED THRESHOLD, ORIGINAL (UNCLIPPED) VS. CLIPPED PIPELINE. ANOMALOUS SLICES REPORT RECALL; BENIGN SLICES REPORT FPR. CLIPPING RAISES OUT-OF-RANGE ANOMALY RECALL ON WADI ($0.31 \rightarrow 0.46$) AND REDUCES IT THREE POINTS ON SWaT ($0.85 \rightarrow 0.82$); WE OBSERVE NO MAJOR RECALL LOSS HERE BUT DO NOT CLAIM CLIPPING PRESERVES ANOMALY EVIDENCE GENERALLY.

Data	Slice	n	orig.	clipped
WADI	benign, in-range	6664	0.000	0.009
	benign, out-of-range	9515	0.085	0.079
	anom., in-range	302	0.000	0.007
	anom., out-of-range	699	0.313	0.464
SWaT	benign, in-range	36651	0.045	0.045
	benign, out-of-range	2747	0.111	0.120
	anom., in-range	1476	0.173	0.165
	anom., out-of-range	4017	0.853	0.824

TABLE S14

DETECTOR-LEVEL REPRODUCTION: IPAL F1 UNDER BETA ON THE ORIGINAL (UNDEFENDED) PATH VERSUS THE CORRECTED SYMMETRIC PATH (GDN, F1-OPTIMAL THRESHOLD). ORIGINAL-PATH VALUES ARE THE SEED-MATCHED RAW DETECTOR; SYMMETRIC-PATH VALUES ARE THE 5-SEED MEANS OF THE MAIN-TEXT DEFENDED-F1 TABLE. BETA'S F1 IMPACT IS FAR BELOW WHAT ITS BETA-REPORTED $\approx 18,900$ -UNIT TARGET-SCORE DEGRADATION IMPLIES.

Data	Metric	Original path		Symmetric path	
		Clean	BETA	Clean	BETA
GDN-WADI	F1	0.28	0.21	0.380	0.341
GDN-SWaT	F1	0.77	0.77	0.657	0.658